

(a) Theoretical limitations

1. Exactly what CPU/GPU do you have in your own computer? What do you know about this specific CPU/GPU and its capabilities?

My computer is equipped with a hybrid computing architecture featuring both a high-performance processor and a dedicated graphics card. The CPU is a 13th Generation Intel Core i7-13700H (Raptor Lake architecture). It utilizes a big.LITTLE core design, containing 14 physical cores (6 Performance-cores and 8 Efficient-cores) and 20 logical threads^[1]. Crucially for parallel computing, its P-cores support advanced vector extensions like AVX2 and FMA (Fused Multiply-Add) instructions, allowing multiple floating-point operations to be executed simultaneously. On the graphics side, in addition to the integrated Intel Iris Xe GPU, the system houses a dedicated NVIDIA GeForce RTX 4060 Laptop GPU (Ada Lovelace architecture, AD107 chip) with 8.0 GB of GDDR6 VRAM. This GPU is equipped with 3072 CUDA cores explicitly designed for massive data-parallel tasks^{[3][4]}, making it vastly superior to the CPU for highly parallelized floating-point arithmetic.

2. What is the clock frequency at which it should run?

The clock frequencies of modern mobile processors are highly dynamic and depend on thermal limits and workload. For the CPU (i7-13700H), the base clock frequency is specified at 2.40 GHz. However, under load, it utilizes Intel Turbo Boost technology; the P-cores can reach a maximum turbo frequency of 5.00 GHz, while the E-cores can boost up to 3.70 GHz^[1]. During my baseline observations in the Windows Task Manager, the CPU actively fluctuated around 4.72 GHz under a moderate load. For the GPU (RTX 4060 Mobile), the frequency is dictated by the laptop's Total Graphics Power (TGP) configuration. While base clocks typically range from 1.47 GHz to 1.89 GHz, the boost clock can dynamically reach up to approximately 2.37 GHz when thermal conditions permit^[3].

3. many floating-point operations should it be able to perform at best per second if you calculate it based on what you know... Please give detailed pointers to the documentation.

To calculate the theoretical peak performance in Single Precision (FP32, corresponding to the float type used in our course), we use the general formula:

$$\text{Theoretical FLOPS} = \text{Cores} \times \text{Clock Frequency} \times (\text{FLOPS/Cycle})$$

For the CPU, a **How** P-core with AVX2 can execute two 256-bit FMA instructions per cycle (processing 8 floats * 2 operations = 16 FLOPs per FMA, totaling 32

FLOPs/cycle)^[2]. An E-core typically executes 16 FLOPs/cycle. Using their maximum boost clocks for the absolute theoretical peak:

$$\text{CPU Peak FLOPS} = (6 \times 5.0 \times 10^9 \times 32) + (8 \times 3.7 \times 10^9 \times 16) = 1.4336 \times 10^{12} \text{ FLOPS}$$

This yields approximately 1.43 TFLOPS for the CPU.

For the GPU, the RTX 4060 has 3072 CUDA cores, each capable of 2 FLOPs per cycle (via FMA)^{[4][5]}. Using a standard maximum boost clock of 2.37 GHz:

$$\text{GPU Peak FLOPS} = 3072 \times 2.37 \times 10^9 \times 2 = 1.4561 \times 10^{13} \text{ FLOPS}$$

This yields approximately 14.56 TFLOPS, demonstrating a 10x theoretical advantage over the CPU.

(Note: Detailed pointers to the architectural manuals and specifications used for these calculations can be found in the references below. Copies of relevant documents or active links are also organized in the `source_materials` folder within my GitHub repository.)

(b) Practical limitations

1. Benchmark Implementation and Operations Calculation

To measure the practical floating-point limits of my hardware, I wrote a C++ program (benchmark.cc) that performs a massive, continuous stream of arithmetic operations. The program takes a float input to prevent the compiler from evaluating the results at compile-time. To ensure maximum execution port saturation, I utilized OpenMP to spawn 20 threads (matching my logical core count) and employed AVX2 intrinsic functions (`_mm256_fmadd_ps`) with a manual loop unrolling factor of 12. Each 256-bit Fused Multiply-Add (FMA) instruction processes 8 single-precision floats and performs 2 operations (multiply and add) simultaneously, equating to 16 FLOPs per instruction. The exact calculation for the total executed floating-point operations is:

$$1,500,000,000 \times 12 \text{ (FMAs)} \times 16 \text{ (FLOPs)} \times 20 \text{ (threads)} = 5,760,000,000,000 \text{ FLOPs}$$

This equals exactly 5.76 TeraFLOPs executed during the run.

2. Compilation and Execution Measurement

The program was compiled in a Linux environment (WSL) using the GCC compiler with aggressive optimization flags: `g++ -O3 -march=native -fopenmp benchmark.cc -o benchmark`. The `-march=native` flag is critical here, as it allows the compiler to fully utilize the specific AVX2 and FMA hardware capabilities of the Raptor Lake architecture. Using `std::chrono::high_resolution_clock` to measure the precise wall-clock time, the program took exactly 5.18918 seconds to complete, providing a

reliable and sufficiently long measurement window.

3. Hardware Telemetry Gathered During Execution

During the benchmark's execution, I closely monitored the system telemetry. The CPU utilization instantly spiked to and maintained a flat 100% across all logical processors. Notably, while the absolute theoretical peak of the P-cores is 5.0 GHz, the sustained clock frequency under this extreme AVX2 workload stabilized at 4.52 GHz due to the laptop's thermal and power limits (TDP).

4. Performance Calculation and Discrepancy Analysis

By dividing the total executed FLOPs by the wall-clock time, my program achieved a sustained practical performance of:

$$5,760 \text{ GFLOPs} / 5.04137 \text{ s} \approx 1.11 \text{ TFLOPS}$$

Compared to the absolute theoretical peak of ~ 1.43 TFLOPS calculated in part (a), this practical benchmark achieved roughly 80% of the theoretical maximum.

Reaching this level of efficiency on a mobile processor is an exceptional result that effectively represents the practical performance ceiling for this specific hardware. As the project guidelines suggest, it is physically impossible to close this gap further due to compounding hardware realities.

Primary among these limitations is the effect of thermal throttling and inherent AVX offsets. The baseline theoretical calculation unrealistically assumed that all Performance-cores (P-cores) could sustain a maximum 5.0 GHz boost indefinitely. In reality, telemetry data confirmed that the CPU throttled heavily to 4.37 GHz when executing dense 256-bit FMA instructions in order to remain within its thermal design power (TDP) envelope. This reduction in clock speed alone accounts for an approximate 12.6% loss in theoretical performance, explaining the vast majority of the empirical discrepancy.

Furthermore, this thermal limitation is exacerbated by the asymmetric nature of the big.LITTLE architecture. The Efficient-cores (E-cores) inherently lack the same raw FMA execution throughput as the P-cores. Because the OpenMP implementation relies on barrier synchronization at the end of the parallel computing region, the faster P-cores inevitably spend a fraction of their time idling while waiting for the slower E-cores to complete their assigned workload. Finally, hyper-threading introduces unavoidable micro-architectural overhead. In this environment, 12 of the 20 threads must share the physical execution ports of the 6 P-cores. Although hyper-threading optimizes overall task scheduling, it cannot physically double the FMA throughput, leading to slight but cumulative instruction queuing. Ultimately, these thermal, architectural, and scheduling constraints collectively account for the remaining variance between the theoretical model and the practical benchmark.

5. Assembly Code Analysis

To verify that the compiler generated the exact machine-language instructions I intended, I output the assembly code using the -S flag (benchmark.s). Upon inspecting the core .L2 loop, I found the following sequence:

```
.L2:
    vfmadd132ps    %ymm0, %ymm0, %ymm6
    vfmadd132ps    %ymm0, %ymm0, %ymm7
    subq $1, %rax
    vfmadd132ps    %ymm0, %ymm0, %ymm3
    vfmadd132ps    %ymm0, %ymm0, %ymm8
    vfmadd132ps    %ymm0, %ymm0, %ymm4
    vfmadd132ps    %ymm0, %ymm0, %ymm12
    vfmadd132ps    %ymm0, %ymm0, %ymm2
    vfmadd132ps    %ymm0, %ymm0, %ymm9
    vfmadd132ps    %ymm0, %ymm0, %ymm5
    vfmadd132ps    %ymm0, %ymm0, %ymm10
    vfmadd132ps    %ymm0, %ymm0, %ymm11
    vfmadd132ps    %ymm0, %ymm0, %ymm1
```

This confirms that the compiler successfully mapped the C++ intrinsics directly to purely vectorized vfmadd132ps instructions operating on 256-bit %ymm registers. Crucially, there are no memory load/store instructions (like vmovaps) inside the loop body, proving that the data remained strictly inside the CPU registers. The loop control was reduced to a simple subq and conditional jump. This validates that the software was optimally compiled, and the 1.14 TFLOPS result represents the true physical thermal limit of the processor.

(c) Comparison with your CP solutions

1. CP Performance Measurement and FLOPs Calculation

To evaluate the synthetic limits against a real-world application, I analyzed the performance of my fastest solution for the CP9a exercise, which computes a correlation matrix using improved algorithmic techniques such as Strassen's algorithm. According to the benchmarks/4 parameters, the input consists of $n_y = 9000$ vectors, each containing $n_x = 9000$ elements. Since the correlation matrix is symmetric, the program only computes the upper triangle, resulting in the following calculation.

$$9000 \times (9000-1)/2 = 40,495,500 \text{ unique vector pairs}$$

Assuming the vectors are pre-normalized, calculating the correlation of each pair requires a dot product of 9000 elements. This mandates 9000 multiplications and roughly 8999 additions, totaling approximately 18,000 floating-point operations per pair. Consequently, the total arithmetic workload executed by CP9a is calculated as follows.

$$40,495,500 \text{ pairs} \times 18,000 \text{ FLOPs/pairs} \approx 7.289 \times 10^{11} \text{ FLOPs}$$

When benchmarking this code locally on my personal machine using the Windows Subsystem for Linux environment, the execution time for benchmarks/4 was recorded at 2.145 seconds. Therefore, the practical floating-point performance of the CP9a solution is derived by the following equation.

$$7.289 \times 10^{11} \text{ FLOPs} / 2.145 \text{ s} \approx 339.8 \text{ GFLOPs}$$

(Note: For thoroughness, I also compared this against the grading server's hardware, which completed the same task in 1.435 seconds, yielding ~0.51 TFLOPS. The slower local time is primarily due to the virtualization overhead of WSL and the strict thermal constraints of a laptop chassis compared to a dedicated cooling environment in a server.)

2. Comparison and Explanation of Differences

An analysis across the three stages of this research project reveals a clear hierarchy of performance degradation as theoretical limits encounter physical and architectural constraints. The theoretical peak established in part (a) is approximately 1.43 TFLOPS. The synthetic practical ceiling measured in part (b) reached 1.14 TFLOPS. Finally, the real-world performance of CP9a on the local machine achieved 0.34 TFLOPS.

The highly optimized CP9a solution achieves roughly 29.8% of the synthetic practical limit of 1.14 TFLOPS. Although 0.34 TFLOPS remains a highly efficient result for dense matrix operations, the significant performance reduction from the synthetic ceiling is an expected outcome that serves as a practical demonstration of the Memory Wall concept in parallel computing.

In the synthetic benchmark of part (b), all arithmetic operations were explicitly unrolled and strictly confined to 12 AVX registers. This methodology effectively bypassed the memory hierarchy to eliminate memory latency and keep the FMA execution ports continuously active. Conversely, the CP9a algorithm cannot maintain all working data within the CPU registers. Calculating massive cross-correlations necessitates constantly fetching arrays of 9000 floats from the main memory or lower-level caches into the L1 cache. Even with advanced cache-blocking techniques and Strassen's algorithm reducing the asymptotic complexity, the arithmetic logic units of the CPU inevitably remain idle for numerous clock cycles while waiting for memory load instructions to retrieve data.

The impact of this bottleneck can be demonstrated through a theoretical modification. If the code were altered so that all matrix memory lookups resulted in an immediate L1 cache hit, thereby simulating infinite bandwidth and zero latency, the memory wait times would be eliminated. Under these hypothetical conditions, the execution units would process the FMA instructions sequentially without interruption, allowing the performance of CP9a to closely approach the 1.14 TFLOPS synthetic limit.

Furthermore, real-world algorithms such as CP9a inherently require loop index arithmetic, branching, and memory management overhead. These factors further

dilute the pure floating-point throughput compared to the uninterrupted stream of vector mathematics executed in part (b).

Reference

- [1] Intel Corporation. (2023). *Intel® Core™ i7-13700H Processor Specifications*. Intel ARK (Automated Relational Knowledgebase). Retrieved from <https://ark.intel.com/content/www/us/en/ark/products/232125/intel-core-i713700h-processor-24m-cache-up-to-5-00-ghz.html>
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- [3] TechPowerUp. (2023). *NVIDIA GeForce RTX 4060 Mobile GPU Specifications*. TechPowerUp GPU Database. Retrieved from <https://www.techpowerup.com/gpu-specs/geforce-rtx-4060-mobile.c3946>
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